

Evaluate the internal consistency of Epicurean metaphysics and whether the swerve is a satisfactory explanation for the formation of compound bodies.

The swerve of atoms in Epicurean metaphysics first appears to be an ad-hoc defence against determinism, but upon an examination of the Epicurean picture of the early universe¹, it proves to be a necessary feature of the theory to explain the formation of compound bodies. This essay first assesses the physical necessity of swerve and then proposes an amendment to it to increase the internal consistency of the theory.

All references of this form (e.g. Epicurus, 5A; Lucretius 5B) are to the passage number in the course handout: (Tor, 2025)

Foundational Metaphysics

The Epicureans held that everything which exists must either be tangible substance or intangible substance. Since the intangible substance cannot provide any resistance to the tangible substance, the tangible substance can occupy it. We can call an occupied region of intangible substance ‘place’, and ‘void’ when it is unoccupied. The things which would be occupying, parts of the tangible substance, can also be called ‘bodies’. Despite being intangible, the void must have extension as it must cover the entire universe to allow atoms to occupy that room (Epicurus, 5A; Lucretius, 5B).

Epicurus argues that the universe is infinite by rejecting the idea that the universe can have an ‘extremity...viewed as next to some further thing’ (Epicurus, 10A). If the universe were finite, it would have an extremity. If the universe were to have an extremity, it could not possibly be next to anything since not even the void extends beyond it. The extremity of the universe would not be so objectionable if we merely considered it from inside the universe. This is because we could view one extremity of the universe as being opposite to that extremity which happens to be on the other side of the universe. We are inside the universe and we cannot escape it since there is no void beyond it for us to occupy. Since there is no void, not even a position for us to define the extremity of the universe, we could just accept that the extremity of the universe can only be defined from inside it. Whether the size of the universe is finite or infinite, it cannot have an extremity from beyond it. Our difficulty in perceiving the extremity of the universe from beyond it does not depend on whether we can figure out the exact point where the universe begins, rather it depends on our inability to observe it from a particular point from outside the universe. This is not an instance of the continuum fallacy as it is an inability to measure in the first place. Since we have not been able to determine whether the universe is finite or infinite in size, we will have to consider both cases.

The void is also recognised as being critical to the destructibility of bodies. Epicurus accepts that ‘of bodies some are compounds, others the constituents of those compounds’ (Epicurus, 8A). However, he demands that there must be some fundamental constituent parts of those bodies, which are ‘incapable of dissolution at any

¹ Given the Epicurean view of the universe being infinite in time, the ‘early universe’ cannot exist. This term specifically refers to the universe before the first swerve – which cannot exist if there are infinite swerves in time.

point or in any way' because 'all things [are not] destroyed into the non-existent but [are] strong enough to survive the dissolution of the compounds'. So compound bodies must be composed of some fundamental parts which are indestructible. Lucretius argues for the indestructibility of the atoms by first postulating that 'the first bodies are solid and without void' (Lucretius, 8B I). Although it may seem unclear at first whether Lucretius is precisely referring to atoms or potentially also some compound bodies which may have also first existed, it becomes clear that he is referring only to atoms, given the picture of the early universe which the Epicureans hold to – which will be explained later in this essay. Accepting the lack of void in the atom is a very important first step for Lucretius because that is how objects experience change. Initially it appears reasonable that 'without void nothing can either be crushed, broken or cut in two'. For a body to be crushed, all the constituent atoms must be brought together so that the void between them is less than it was before. Therefore, something cannot be compressed if it has no void to travel. This explanation from the amount of void present in some object, however, is not sufficient to explain why an atom could not be sliced in two. For example, if an atom is stationary and another atom collides with it at an angle, there is no lack of void which would prevent that part of the stationary atom which was hit to simply slide off the rest of it. For a body composed of many atoms, the Epicureans do have an explanation for why some objects cannot simply be pulled apart without much effort. Epicurus speaks about the 'shapes, colours, sizes, [and] weights of bodies (Epicurus, 7B), as though there are many possible shapes an atom can take, but he does not refer to atoms explicitly. Lucretius continues to write in detail about how the motions and densities of the primary bodies determine the compound body which the atoms ultimately form such as the atoms which 'recoil and rebound a long way over great intervals' supplying us with sunlight. In the same passage, he also mentions that atoms which are 'in a denser aggregation and rebound over small intervals from impact' are 'held back by the interweaving of their own shapes'. We may ask ourselves how it is possible that perfectly spherical atoms can be interwoven, but the Epicureans were never committed to that shape. So, one reason why it may be difficult to pull apart a compound body such as a rock is because constituent atoms have hooks just as indestructible as the rest of their shape which are pulling onto each other.

Although it is a commendable effort to try to remain as self-consistent as possible by referring to other parts of one's metaphysics rather than always introducing new ideas, in this case, Lucretius' argument that atoms are indestructible because they contain no void is insufficient. Epicurus's argument is less mechanical and so perhaps not as satisfying as Lucretius'. We may say that Epicurus cannot rule out the existence of a minority of atoms which may be destroyed after a certain number of dissolutions of their compound body, or perhaps some atoms which can only exist for a limited amount of time. For example, a very small amount of such atoms might be present in some sugar and then disappear when that sugar is dissolved in water. But if we take Epicurus's thinking and we accept that there is even a single atom which is indestructible, then it must have always existed and it will continue to always exist – so the universe must be infinite in time. Since Epicurus holds that these atoms exist in a universe which is infinite in time, we can be certain that such 'temporary atoms' would have already been destroyed infinitely long ago. Therefore, we could only possibly see the remaining atoms which are truly infinite and can survive any kind of dissolution and any period of time. It remains possible that no such indestructible atom exists and therefore that the universe has no reason for being infinite.

There is, however, a problem with the existence of only a single atom existing at any point in time. If there is only one single atom in a universe which is infinitely large, no possible change could be detected in its position or velocity because it cannot be observed moving relative to any position we can fix in space. Even if there is a single atom and a single 'temporary atom', and they happen to be moving in exactly the same direction and at the same speed, no possible change can be detected in the system so there is no time. The atoms cannot even measure their movement relative to any extremity of the universe because the universe is infinitely large. Therefore, it is possible to have infinitely many bodies and void, as Epicurus argues (Epicurus, 10A), and yet for time to be finite. One may argue that a succession of positions must occur in time, whether it can be observed or not, but if we are unable to distinguish between two positions because we cannot observe any difference at those points, we are equally justified in saying we have moved just as much as we would be in saying that have not. Therefore, we can argue that Epicurus has overestimated how much time has passed as it may not necessarily be infinite depending on the behaviour of the atoms. According to Epicurus, nothing beyond body and void can be imagined 'as completely substantial things and not as things which we call accidents or properties of these' (Epicurus, 5A). But since we can imagine time, it must therefore be by a kind of property of a body. This property must exist whether it can be detected. So far the Epicureans, detection of change has no consequence of the passage of time. The Epicurean picture of the early universe is precisely one in which time is undetectable for infinitely many atoms which are 'falling down like raindrops through the depths of the void' (Lucretius, 11H). One may try to defend this picture of every atom falling at the same speed by hoping that it may be an explanation of gravity. But since the Earth itself and everything on it is also made of atoms, the Earth and everything on it must also be falling down at the same rate – meaning that anything thrown upwards would never come down again from our perspective.

So now we have a picture of the early universe in which the atoms all have some undetectable 'downwards' motion but no lateral motion. The Epicureans have specified a certain picture of the early universe for which time could not be observed. However, even though the changes in the positions of every atom cannot be observed, the Epicureans can remain certain of it because they assert that all atoms have weight which causes them to fall in the void. Therefore, there must be time, although it cannot be detected. Since the void offers no resistance to the downwards or lateral motions of the atoms, they do not exhibit the Brownian motion that we observe today and they all fall down at an undetectable speed. The Epicureans introduce the swerve of atoms to explain how they began to form us and the other compound bodies with which we interact.

Introducing Swerve

Lucretius stated that the atoms 'at quite uncertain times and places...veer a little from their course' (Lucretius, 11H). Swerve is a necessary feature of Epicurean metaphysics because there is nothing previously examined that could permit for lateral movement of the atoms. Epicurus does not list swerve as a 'permanent attribute' of a body alongside 'shapes, colours, sizes, [and] weights' (Epicurus, 7B). This is because it is not essential to the atom. Swerve is a kind of attribute or a capacity of the atom, but if the atom happens to not be swerving at a particular point in time, it does not cease to be that same atom. The atom's property of time is similar

because even though it is always in motion and therefore always in time, being in time is not essential to the atom.

Accepting Swerve

But in permitting this ‘uncertain’ swerve, the Epicureans lose the hard chain of causation which explains the development of the universe. The Epicureans could have conceded to a single, uncaused swerve but they choose to keep it as a capability of every atom which is exercised at ‘uncertain times’. In doing so, they concede more uncaused events into their system when they are no longer necessary as only one swerve would be necessary to initiate the Brownian motion which we are looking for. One possible objection may be that, over the span of an infinitely large universe, one single swerve would not be sufficient for the disorder of Brownian motion to catch up to the rest of the universe and so the region of the universe that looks like ours is relatively small – but we require that the entire universe is not just falling straight down but that it has developed and formed compound bodies. So now, the requirement that the universe is infinitely large forces the Epicureans to admit infinitely many uncaused events into their universe. We may reject the requirement that every part of the universe must look like ours and that it is acceptable for some parts of the universe to have no disorder at all, as Epicurus describes ‘an infinite number of worlds’ which are ‘similar to this one and of those which are dissimilar’ (Epicurus, 13A). But a dissimilar world gives the impression of still being a world with disorder compared to the perfectly ordered ‘raindrops through the depths of the void’. Therefore, we require that the swerve is occurring over all space.

Furthermore, accepting the swerve as a single exception to the perfect determinism of the universe raises the problem of its ontological status. Since nothing can even be conceived of which is neither body nor void, besides ‘accidents or properties of these’ (Epicurus, 5A), this singular swerve must be a property of either a body or a region of void. It cannot be a property of the void since the void itself cannot swerve, only atoms can. But if it is a property of an atom, could it be a property of only a single atom? Although it seems to make the theory less internally consistent, it is not impossible. If we consider that swerve is not a permanent attribute but rather a capacity which is only sometimes exercised, we could still accept that every atom has the capacity to swerve but that only one ever did. It seems more consistent to accept swerve as a capacity of every atom than to have a perfectly deterministic universe with a single atom as an exception. But we would be unfortunately positing that every atom, with one exception, has a completely unobservable capacity – this seems to depart from Epicureanism’s strength of having a strict ontology which has so far has posited no more than what has been deemed philosophically necessary. It would be more consistent to permit all atoms to exercise this capacity or to not let any atoms have it in the first place.

Modifying Swerve

Having accepted that it is not possible to limit the extent of swerve in space, this paper will examine whether it may be restricted in time. The requirement for atoms swerving not just once but during the entire duration of the universe comes from Lucretius’ mention that ‘at uncertain times and places they veer a little from their course’. If the swerve of an atom is truly uncertain, then it may certainly happen more than once and we cannot rule out that it will never happen again. But if we modify the theory such that atoms may only swerve once, we can reduce the number of uncaused events in the universe while still being able to explain Brownian

motion observed across an infinite universe. We can simply change the capability of the atom to swerve from being uncertain but always possible to being uncertain but possible only once. This would provide us with the first cause of motion, across the entirety of infinite space, while also no longer maintaining it when it is not required in the future after some particular region of space has already experienced lateral motion. Not only that, but we still have the same level of consistency as we had before because we have not introduced a new law - we have simply modified a previously existing law. In fact, we have made it simpler by reducing it from uncertain and eternal swerve to a one-time swerve for every atom. As more atoms exhaust their one-time swerve in a particular region, Brownian motion takes over as the most significant effect for the lateral motion of the atoms instead of the swerve which caused it.

One may argue that limiting the swerve of each atom to once leaves the theory vulnerable as it now must address a first, singular uncaused event. If there is no first swerve because they occur infinitely in time, then there is no first uncaused event. But this is only the case if the swerves are causally linked. No two swerves can be causally linked because every swerve is uncaused. Since they are not causally linked, every single swerve is an instance of an uncaused effect and increasing the number of swerves only worsens the problem. Making the number of swerves infinite in time by increasing the number of swerves per atom is also unnecessary because it is already contained within the infinity of the atoms in space. There are an infinite number of atoms in space, since they will not all swerve at once, there are enough atoms to swerve at every moment in infinite time – if it were necessary.

In conclusion, Epicurean metaphysics is a generally consistent theory which only accepts that which it has recognised as being necessary. Swerve, given the previously established metaphysics, is a necessary concession to account for the formation of compound bodies. This essay has sought to increase the internal consistency of the theory by reducing the number of swerves while preserving the system as envisioned by the Epicureans. Unfortunately, the number of swerves, despite being limited at one per atom, remains infinite in the universe due to the infinite number of atoms in space. Reducing the number of swerves per atom did limit the infinity of swerves per atom in time – but the number of swerves in the universe remains infinite. While the theory has been improved upon, the number of uncaused events remains extremely high. If, however, we reject the conclusion of Epicurus's argument for the size of the universe being infinite, we now no longer need infinitely many atoms to occupy an infinitely large universe. Therefore, we can guarantee a finite number of swerve as we have blocked the infinite swerves in time per atom and the infinite number of atoms.

Bibliography

Tor, S. (2025). *Week 5: Epicurean physics: atoms, void & freedom (Course handout)*. London: King's College London, Department of Philosophy.